

Taewoo Kim

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RESEARCH INTEREST

- **Primary Focus:** Electrode-electrolyte interface engineering and materials design for Li- and Na-based batteries
- **Key Approaches:** Designing novel active materials and electrolytes; investigating solid-state battery interfaces; and integrating experimental characterization with computational modeling.

EDUCATION

KAIST - Korea Advanced Institute of Science and Technology Mar. 2019 - Aug. 2026 (Expected)
Undergraduate Program Daejeon, South Korea

- Primary Major — **Chemical and Biomolecular Engineering**
- Double Major — **Materials Science and Engineering**
- Overall GPA: **4.21/4.3** (3.99 on 4.0 scale) ; ChBE GPA: 4.19/4.3, MSE GPA: 4.3/4.3
- Leave of Absence (Fall 2022 - Spring 2024) due to the mandatory military service

Georgia Institute of Technology Aug. 2024 - May 2025
Exchange Program in Chemical & Biomolecular Engineering Atlanta, Georgia, U.S.A.

- Overall GPA: **4.0/4.0**

RESEARCH EXPERIENCE

Energy Materials Lab — PI: Prof. Nam-Soon Choi Jul. 2025 - Present
Undergraduate Research Assistant Dept. of CBE, KAIST

- **Project: Designing Electrolyte Suitable for Na-ion Batteries under Low-Temperature Conditions**
 - Investigated the fundamental properties of electrolytes and current research trends through various literature
 - Established key experimental conditions by screening several types of additives and optimizing the N/P ratio for the full cell testing

Liu Resarch Group — PI: Prof. Meilin Liu Aug. 2024 - May 2025
Undergraduate Research Assistant School of MSE, Georgia Tech

- **Project: Investigating Effect of Surface Treatment on Ni/Mn-based Co-free Layered Cathode for Li-ion Batteries**
 - Demonstrated significant performance enhancements by applying a lithium-molybdenum-phosphate (LMP) surface treatment using planetary mixing and calcination
 - Conducted various electrochemical tests to assess the performance of the fabricated cells
 - Analyzed the data acquired from the microscopic characterizations to understand the coating chemistry
 - Shared progress with the group members through bi-weekly meetings and technical presentations

Electrochemical Energy Devices Lab — PI: Prof. Hee-Tak Kim Jan. 2021 - Dec. 2021
Undergraduate Research Assistant Dept. of CBE, KAIST

- **Project 1: Optimizing Composition of Cathode Materials in Flowless Zn-Br Aqueous Battery**
Individual Research (Jan. 2021 - Jun. 2021)
 - Found optimal composition of Super P & Activated carbon through galvanostatic cycling test
 - Employed various types of ordered mesoporous carbons to increase surface area
 - Employed h-TiN catalyst to improve capturing of bromine ions and enhance reactivity
- **Project 2: Identifying Key Operation Conditions of Electrolytic Zn-MnO₂ Battery**
URP - Undergraduate Research Participation Program (Jul. 2021 - Dec. 2021)
 - Found optimal operational current of the system through rate capability test
 - Examined underlying mechanism of cell degradation through cyclic voltammetry and EIS
 - Enhanced long-term energy efficiency by introducing an electrolyte reservoir to maintain the pH

PUBLICATION

1. Y. Ahn, X. Hu, Y. Ding, C. Kim, Y.-C. Wu, **T. Kim**, M. Liu*, **Synergistic Effects of Entropy Tuning in Niobium-Based Oxide Anode for Fast-Charging Lithium-Ion Batteries.** *Adv. Funct. Mater.* 2025, e09533. <https://doi.org/10.1002/adfm.202509533>

TECHNICAL SKILL

Synthesis & Fabrication	Solid-state Synthesis, Slurry Casting, Coin Cell Assembly
Electrochemical Analysis	Galvanostatic Cycling, Cyclic Voltammetry, EIS, GITT
Materials Characterization	<i>Hands-on:</i> XRD; <i>Data Analysis:</i> XPS, Raman, SEM
Software & Programming	Origin, Python (ML techniques), MATLAB (Molecular simulations), L ^A T _E X

ACADEMIC PRESENTATION & AWARD

Georgia Tech 19th UROP Spring Symposium	Apr. 2025
<i>Poster presentation</i>	Undergraduate Research Opportunities Program, Georgia Tech
Georgia Tech 12th Annual MRS Poster Competition	Apr. 2025
<i>Poster presentation</i>	Materials Research Society (MRS) @ Georgia Tech
· Title: Effect of Lithium-Molybdenum-Phosphate Surface Coatings on Ni/Mn-Based Co-Free Layered Cathode for Lithium-ion Batteries (Link)	
KPF Student Academic Conference - Best Study Award	Feb. 2022
<i>Third Prize out of 26 teams</i>	KPF Winter Workshop, KAIST
· Title: Developing Electrolytic Zn-Mn Battery System and Identifying Key Operation Conditions (Link)	

HONOR & SCHOLARSHIP

KAIST Presidential Fellowship (KPF)	Mar. 2021 - Present
<i>Selected for having outstanding creativity and potential among undergraduates</i>	Global Leadership Center, KAIST
National Presidential Science Scholarship	Mar. 2021 - Dec. 2024
<i>Awarded to Korean students having outstanding potential in STEM field</i>	Korea Student Aid Foundation
KOR-US High-Tech Industry Student Exchange Scholarship	Apr. 2024
<i>Part of KOR-US STEM Educational Exchange Initiative</i>	Korea Institute for Advancement of Technology
Academic Excellence Scholarship	
<i>Merit-based scholarship awarded to top 4 students in GPA</i>	Dept. of CBE, KAIST
· Awarded semesters — Fall 2020 (2 nd place), Spring 2021 (1 st place), Fall 2021 (1 st place)	
Dean's List	
<i>Top 3% students in the College of Engineering</i>	College of Engineering, KAIST
· Awarded semesters — Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022	

EXTRACURRICULAR ACTIVITY

International House	Spring 2025
<i>Living & learning community shared between GT degree-seeking and exchange students</i>	Georgia Tech
· Engaged with diverse cultures and strengthened global communication skills	
Calculus II Tutor for Freshmen	Spring 2020 - Spring 2022
<i>Delivered short lectures and provided tutoring for 5 consecutive semesters</i>	School of Freshman, KAIST
CBE Student Council	Fall 2020 - Fall 2021
<i>Member in the Division of Culture & Event</i>	Dept. of CBE, KAIST
IT Volunteer - Adobe Photoshop Instructor	Nov. 2020
<i>Conducted in online due to the COVID-19 pandemic</i>	World Friends Korea